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Optimization of reliable vehicle routing problem for medical waste collection with time windows in stochastic transportation networks

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ABSTRACT

The collection and transportation of medical waste (MW) is critical because of environmental and public safety implications. However, the variability in travel times of waste transport vehicles, influenced by various factors, complicates this process. This article proposes a multi-objective optimization model to enhance travel time reliability (TTR) while minimizing vehicle transportation costs and penalties for time window violations. Given the NP-hard nature of the problem, a genetic algorithm enhanced by an improved harmony search genetic algorithm (IHSGA) is utilized. The model and algorithm are evaluated in N City, China. The results highlight the significance of TTR in optimizing MW collection operations. Sensitivity analysis of key parameters validates the model's robustness. Furthermore, comparative analysis demonstrates the superiority of IHSGA over traditional methods. These findings underscore the importance of incorporating additional costs associated with travel time variability in decision making, and advocate for strategies to mitigate such uncertainties.

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Medical waste; uncertainty; travel time reliability; time windows; improved harmony search genetic algorithm (IHSGA)

1. Introduction

Medical waste (MW) is highly infectious, posing vital risks to human well-being and the environment. Therefore, effective medical waste management (MWM) is important in mitigating infection risks, especially concerning coronavirus disease 2019 (COVID-19) waste (Peng *et al.* 2020). With the progression of healthcare reforms and the increasing public awareness of health and environmental protection, there is a growing need to prioritize procedures for MW collection and transfer (Govindan *et al.* 2021). It is important to note that assuming a constant travel time is impractical. Therefore, incorporating uncertainty factors into the design of the entire MW routing network is vital to enhance its practicality. While previous research has advanced the development of more realistic optimization models for MWM, the reliability of travel time remains a factor that has not been fully integrated. For an in-depth examination of this issue, please refer to Section 4, which includes Figure 1, for a comprehensive overview.

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The primary objective of the research is to develop an optimization model that enables the establishment of a safe, sustainable and cost-effective network for MWM. The proposed model considers the transportation aspects of MW and particularly emphasizes travel time reliability (TTR). In summary, this research presents three primary contributions:

- A novel network design optimization model is proposed that incorporates TTR in stochastic road networks, emphasizing uncertainties encountered during the transportation phase.
- A genetic algorithm based on an improved harmony search genetic algorithm (IHSGA) is designed to address the capacitated vehicle routing problems with time windows (CVRPTW). The algorithm offers the advantage of achieving both high solution quality and fast convergence speed.
- The proposed model and algorithm are validated through numerical experiments on a real traffic road network in N City, China. The outcomes of this real-case analysis indicate that policymakers should consider the uncertainties associated with MW transportation, ultimately leading to enhancements in social and economic benefits.

The remainder of this article is organized as follows. Section 2 presents a comprehensive literature review, surveying the existing body of work in the field. Key definitions and concepts that are integral to the study are elucidated in Section 3. Section 4 presents the research content through illustrative examples, providing a tangible context for the theoretical framework. The development of the optimization model is detailed in Section 5, laying the foundation for the subsequent analysis. Section 6 presents the solution methodology used in the study. A case study is presented in Section 7, designed to validate and evaluate the efficacy of the proposed model. Finally, Section 8 summarizes the findings of the study, draws conclusions and proposes potential avenues for future research.

2. Literature review

2.1. Transportation under deterministic conditions

Much of the MWM network planning research has predominantly focused on deterministic scenarios, often overlooking the dynamic variables in real-world applications. In such deterministic contexts, loading capacities and travel times along the routes are presumed to be constant and unvarying. As a result, the body of research conducted within these deterministic frameworks can be effectively divided into two principal areas for review and analysis.

2.1.1. Vehicle routing problem with capacity constraint under deterministic conditions (CVRP-D)

Hannan *et al.* (2018) emphasized optimizing waste collection routes under deterministic routing with capacity constraints. Akhtar *et al.* (2017) extended the CVRP-D to incorporate smart waste bins into this model's characteristics. These studies primarily focus on enhancing waste collection and transportation efficiency and effectiveness. Optimizing routes and considering capacity constraints can lead to reduced costs, minimized environmental impacts and improved waste management systems.

2.1.2. Vehicle routing problem with time windows under deterministic conditions (VRPTW-D)

An iterative algorithm for deterministic time-based path planning is presented in Figliozzi (2010), specifically addressing the vehicle routing problem with time windows (VRPTW). This algorithm was designed to accommodate a range of general costs and to be compatible with improvement algorithms. Kim, Kim, and Sahoo (2006) considered the VRPTW for waste collection for real-life waste collection scenarios. S. Q. Zhang, Mu, and Wang (2020) addressed the collection vehicle routing problem of the garbage facilities (CVRPGF) by reformulating it as a vehicle routing problem (VRP) with simultaneous pick-up and delivery and time windows, by introducing virtual customers.

Building on this work, W. T. Zhang *et al.* (2020) developed a mixed-integer linear programming (MILP) model specifically tailored for optimizing routes in the context of the VRPTW for multi-cycle MWM. Guan *et al.* (2023) proposed a two-level planning model based on a state–spatiotemporal network to optimize the routing of customized buses. They suggested a nested algorithm that combines the GA and dynamic programming algorithm based on augmented Lagrangian relaxation.

Numerous studies have explored different approaches to address this issue and proposed novel solutions, which offer valuable insights for future research directions. For instance, Li *et al.* (2018) adopted reinforcement learning models to model the route and departure time choice behaviour under incomplete information. The study also designed an evolution law for the expected arrival time of passengers and a traffic flow allocation algorithm to assist travellers in avoiding peak times effectively. In another study, Shen *et al.* (2020) proposed a two-stage model for real-time customized bus route design, where column generation was used to solve the proposed model in the first stage (VRPTW). On the other hand, K. Zhang, Lin, and Li (2023) suggested using a reinforcement learning algorithm based on graph attention networks for the multi-depot vehicle routing problem with soft time windows (MD-VRPSTW). Sun, Li, and Li (2022) studied the problem of urban delivery considering order release dates and flexible time windows. The authors propose a MILP model and develop a set of efficient inequalities and branch-and-cut (B&C) algorithms to minimize transportation and delay costs. In addition, Wang *et al.* constructed a cold-chain logistics vehicle routing optimization model and used the Clarke–Wright algorithm to effectively address the 'last kilometre' problem faced by delivery personnel.

In network planning for MWM, diverse research objectives arise from various perspectives. Apart from the conventional capacity-constrained and time-window-constrained variations, other adaptations emerge as a result of differing focal points.

Osaba *et al.* (2019) expanded upon the conventional VRP by integrating additional real-world constraints, redefining pharmaceutical waste collection as a multi-attribute vehicle routing problem (MA-VRP). Taslimi, Batta, and Kwon (2020) tackled the complexities of the periodic load-capacitated VRP, a common challenge for organizations collecting medical and municipal waste. In a related development, Gao *et al.* (2021) merged the VRP tailored for medical facilities with the waste collection needs of clinics, emphasizing the interconnection with their affiliated hospitals. Ghannadpour, Zandieh, and Esmaeili (2021) focused on advancing sustainable waste collection practices at small-scale medical centres in Iran. They pursued this goal by employing various effective strategies to craft an adaptive, multi-objective evolutionary algorithm specifically designed to address the real-world nuances of the VRP in MW collection. In addition, Goli, Ala, and Hajiaghaei-Keshteli (2023) explored some meaningful scenarios, including non-permutation flow-shop scheduling issues and the simultaneous optimization of batch sizes. The research also delves into supply chain design, considering financial and physical flows within the portfolio (Goli and Tirkolaee 2023). Furthermore, integrating these concepts in developing a blockchain-based closed-loop supply chain is a significant endeavour (Goli 2023).

2.2. Transportation under uncertain conditions

In recent years, researchers have increasingly acknowledged the need to integrate uncertainties into their investigations during pandemics such as COVID-19. Previous studies have concentrated on 'demand' and 'travel time' uncertainties. The primary emphasis of this article lies in the uncertainty surrounding travel time. Therefore, from this perspective, the article explores the similarities and differences in existing research.

Asefi *et al.* (2019) dedicated their research to addressing the location routing problem within consolidated solid waste management, particularly under conditions of temporal uncertainty. To confront this computational challenge, they devised a MILP model. In the same vein, Vareias, Repoussis, and Tarantilis (2019) aimed to create a rational allocation scheme for time windows, given the

unpredictable nature of travel times. They introduced a mathematical model incorporating service reliability into the routing process for the VRP with fluctuating travel times. Building on this, they proposed a hierarchical solution strategy to efficiently distribute time windows to customers, tailored to navigate the intricacies of the optimization model that they presented. Furthering the discussion on travel time variability, Zhi, Zhou, and Zhao (2020) developed a method that transforms predicted travel time values into a metric of TTR, using the distributed wave theory to account for the impact of traffic signals on travel times. Ehmke, Campbell, and Urban (2014) examined the stochastic version of the split delivery vehicle routing problem with time windows (SDVRPTW). They approached the problem by assuming that travel times are stochastic, and incorporated probabilistic constraints to mitigate the risk of time window violations for individual deliveries. Q. Wang, Liu, and Pei (2023) proposed a two-layer multi-objective logistics network design model for post-disaster humanitarian relief, which takes into account the uncertainty of transportation time, transportation cost and demand. They develop a globalized, robust two-tier multi-objective model that includes upper-level decision makers focusing on allocation issues and lower-level decision makers considering supply issues. The original model is transformed and solved by the prime–dual method. B. C. Wang *et al.* (2024) studied the problem of resource allocation optimization when multiple secondary disasters occur. The authors propose a two-stage stochastic optimization model to simulate scenarios of random occurrence and severity of multiple natural disasters to optimize the allocation of rescue teams, storage supplies and medical resources. A particle swarm optimization algorithm was used to solve the problem. Goli, Ala, and Mirjalili (2023) devised a probabilistic planning model and a simulation-based solution approach for the placement, allocation and distribution of organ transplant centres to mitigate the uncertainty surrounding delivery times in organ transportation. This initiative led to the identification of optimal credibility levels and improved patient satisfaction.

The existing optimization models increasingly consider realistic factors, including parameters, carbon emissions and medical uncertainty. Günay, Kremer, and Zarindast (2021) devised a robust possibility planning (RPP∗) method to tackle passenger satisfaction and demand uncertainties, using interactive fuzzy programming techniques. Similarly, Habib *et al.* (2021) addressed uncertainties in the biodiesel supply-chain model using an RPP-based solution method, recognizing the substantial uncertainty stemming from the business environment. The challenges of medical equipment shortages and uneven organ distribution were amplified by the COVID-19 pandemic, drawing research focus to these issues. Kargar *et al.* (2020) developed a likelihood planning model that uses pre-emptive goal programming to optimize liver allocation to improve patient survival rates while minimizing transportation costs and time. Alizadeh *et al.* (2024) addressed supply issues related to pulse oximeters and thermometers by proposing a model that prioritizes cost minimization, reliability maximization and enhancing the critical response. Their approach considers greenhouse gas emission uncertainty and uses an efficient algorithm for Benders decomposition. Similarly, Abad, Barzinpour, and Pishvaee (2023) focused on tackling ventilator and infusion pump supply challenges by integrating deep dynamic uncertainty and refining demand characterization. Their research highlights the effectiveness of accounting for deep dynamic uncertainty in reducing model costs.

These studies indeed play a vital role in addressing the challenges related to vehicle routing under travel time uncertainties. To address this issue, they use various techniques, including MILP, metaheuristic methods, and more. Their contributions are significant in enhancing our understanding and ability to deal with the complexities introduced by time uncertainty in path planning.

2.3. Research gap

Previous research has explored integrating the VRP with healthcare waste management. This article, similarly to the existing literature, focuses on the uncertainty surrounding travel time. However, a crucial factor overlooked by current models is TTR, which refers to the probability of completing a trip to specified time windows set by medical units along predefined routes. Furthermore, this study

Table 1. Categorization of current research on networks for managing medical waste.

	Type			Objective function		
Reference	Deterministic	Stochastic	TTR	Vehicle cost	Transportation cost	Time window
Hannan *et al*. (2018)		✓		✓	✓	✓
Akhtar *et al*. (2017)		✓		✓	✓	
Figliozzi (2010)		✓				
Kim, Kim, and Sahoo (2006)		✓				✓
S. Q. Zhang, Mu, and Wang (2020)		✓				
W. T. Zhang *et al*. (2020)		✓			✓	
Osaba *et al*. (2019)		✓				
Taslimi, Batta, and Kwon (2020)		✓				
Gao *et al*. (2021)		✓				✓
Ghannadpour, Zandieh, and Esmaeili (2021)		✓		✓	✓	
Hu *et al*. (2018)	✓					
Lu *et al*. (2021)	✓			✓	✓	✓
Yu *et al*. (2020)	✓					
Asefi *et al*. (2019)	✓			✓	✓	
Vareias, Repoussis, and Tarantilis (2019)	✓					✓
Ehmke, Campbell, and Urban (2014)	✓					
Shen *et al*. (2020)	✓			✓		✓
Guan *et al*. (2023)	✓			✓	✓	✓
Current article		✓	✓	✓	✓	✓

Note: TTR = travel time reliability.

emphasizes heuristic methods to address medium- and large-scale scenarios, providing managers with a practical basis for decision making. Table 1 offers a summary categorization of relevant studies in this domain to assist in navigating the literature.

3. Definitions

- Travel time reliability: TTR corresponds to the probability of meeting predetermined time windows while travelling between medical units along a designated route. It captures the likelihood of successfully navigating the designated path, road or origin–destination (OD) under specific conditions.
- Effective travel time: This concept denotes the specific time intervals within a pre-established confidence level (TTR). The effective travel time is posited to adhere to a logarithmic normal distribution, which is used to model the variability and uncertainty inherent in travel durations. The mathematical formulation encapsulating this definition is as follows:

$$\ln \ln t_{ij}^{\theta} = \mu_{ij}^2 + \Phi^{-1}(\theta)\sigma_{ij}^2 \tag{1}$$

By taking the logarithm of both sides of the equation, Equation (2) can be derived:

$$t_{ij}^{\theta} = exp(\mu_{ij}^2 + \Phi^{-1}(\theta)\sigma_{ij}^2) \tag{2}$$

In Equation (1), μ_{ij}^2 and σ_{ij}^2 represent, respectively, the mean and variance of travel time for link (i, j).

In conclusion, ensuring the definition of TTR entails rationalizing waste collection processes while maintaining carrier capacity and ensuring that TTR meets or exceeds the specified reliability level.

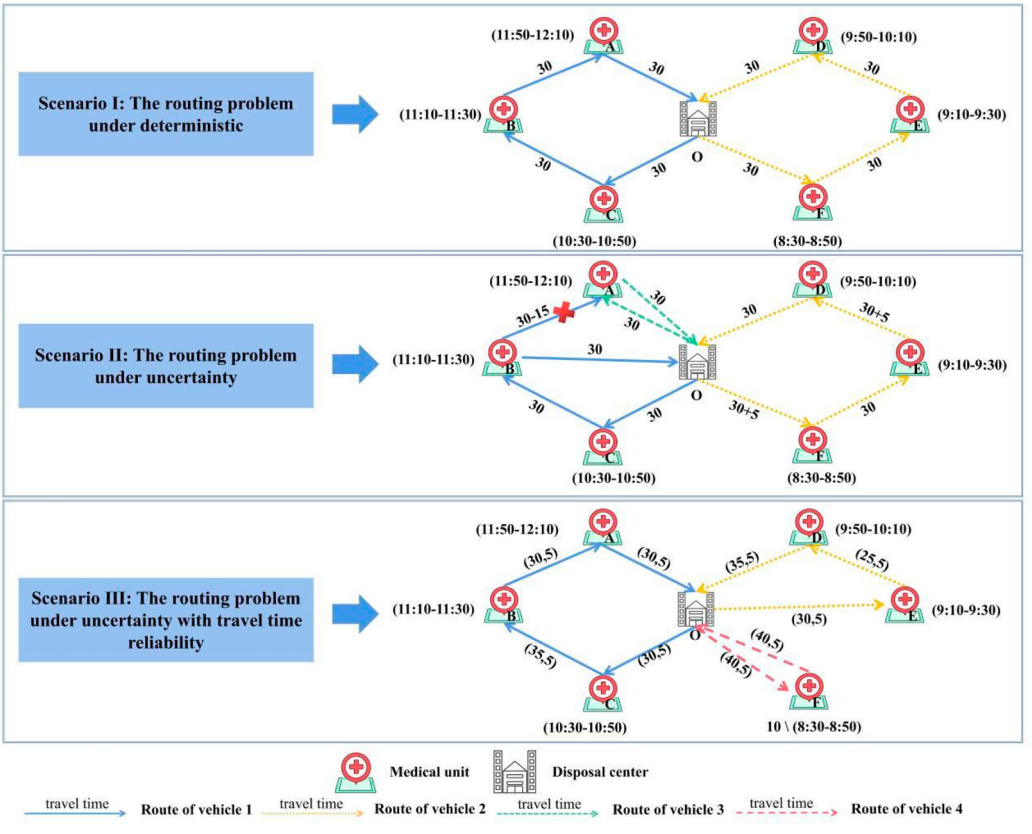


Figure 1. Considering travel time reliability (TTR) under uncertainties.

4. Example

This example examines a compact transportation network of six medical units and one disposal centre, as shown in Figure 1. The figure outlines the routes for MW transportation across three scenarios. For analytical convenience, it is assumed that the disposal centre consistently operates within its designated time window, there are no restrictions on the number or type of vehicles, and each vehicle spends 10 time units at each medical unit.

When travel times are deterministic, route planning aligns with Scenario I, depicted in Figure 1. Each road segment boasts a fixed travel time of 30 units. The specific routes are O–C–B–A–O and O–F–E–D–O. Consequently, vehicles encounter no penalties or delays throughout their journey.

In Scenario II, where uncertainty prevails, variations in travel times are observed across specific road segments. Notably, the travel duration on the segments from O to F and D to E escalates to 35 units, contrasting with a reduction to 15 units on the A to B segment. Adhering to the conventional routes would lead to unnecessary idling and suboptimal utilization of public resources, primarily due to the shortened travel time on the A to B segment. Alternative routes have been strategically designed to mitigate this: O–F–E–D–O, O–C–B–O, and O–A–O. These newly engineered routes are crafted to ensure compliance with time window limitations, thus making this routing strategy prudent.

Next, the concept of reliable routing is introduced, designated as Scenario III, which accounts for the inherent randomness in travel times across different road segments. A normal distribution is used to model the stochastic variables associated with these travel times, with the mean and variance of these variables provided as numerical values in parentheses. This approach aims to provide a

more robust and predictable routing framework, accommodating the variability in travel times and enhancing route planning efficiency.

The travel time of the route can be represented as:

$$\sum_{r_i=1}^{R} \mu_{ij} + \phi^{-1}(\theta) \sqrt{\sum_{(i,j)\in r_i} \sigma_{ij}^2} \tag{3}$$

The TTR is set to 80% ($\phi^{-1}(0.8) = 0.84$).

In Scenario III, where TTR is a key consideration, the strategic routes have been meticulously planned to optimize travel efficiency. The designated routes are as follows: vehicle 1 traverses the path O−−C–B–O with an allocated travel time of $128.76 = (30 + 35 + 30 + 30) + 0.84 \times \sqrt{5 + 5 + 5 + 5}$; vehicle 2 follows the route O–E–D–O, which is projected to take $93.25 = (30 + 25 + 35) + 0.84 \times \sqrt{5 + 5 + 5}$ units of time; and vehicle 3 takes the direct route O–F–O, estimated to require $82.66 = (40 + 40) + 0.84 \times \sqrt{5 + 5}$ units of travel time. Each of these meticulously designed routes is engineered to adhere to the established time window constraints, ensuring a seamless and punctual delivery of services.

It is imperative to acknowledge that while the prevailing route planning strategies prioritize travel time, a comprehensive approach must also integrate other critical elements. In practical transportation contexts, variables such as the risk profile of the transit routes are equally significant. When uncertainty is introduced, the resultant routes can vary significantly from those initially planned. A pronounced divergence exists between the routing strategies that integrate TTR, exemplified in Scenario III, and those that solely consider general uncertainty, as depicted in Scenario II. To provide further clarity, waste disposal facilities often have limited operating hours. Consequently, when managers strive for enhanced reliability, they must navigate the operational constraints imposed by these facilities' time windows. This frequently involves strategically allocating additional vehicles to offset potential time window violations. This pragmatic strategy not only aligns with the practicalities of real-world operations but also serves as a viable solution to the challenges posed by time-sensitive logistics. It underscores the importance of a balanced approach that addresses the temporal and operational facets of transportation planning.

5. Mathematical model

5.1. Problem description

Vehicles frequently encounter challenges accessing medical units throughout the day owing to their engagement in the waste circulation process within the recycling network. This necessitates that vehicles adhere to designated time slots and soft time windows to collect MW. Despite meticulous planning of the transportation routes, unforeseen factors such as inclement weather and traffic incidents can impact the velocity of vehicle travel. Furthermore, there is an inherent lack of certainty that the vehicles will successfully adhere to the time windows stipulated by the medical units. This highlights the complexity of route planning in unpredictable real-world conditions.

Figure 2 shows three situations:

- Successful trip: Vehicle 1 arrives at the medical unit within the designated time window and completes its task to the window's closure. This scenario is deemed a successful transportation trip.
- Early arrival: Upon arriving early, vehicle 2 must wait, since the time window has not yet commenced. Given that medical units use public resources and can impact the flow of other vehicles, vehicles must minimize their duration of stay at these units.
- Delayed arrival: Should vehicle 3 experience a delay, the tight time window may not accommodate the vehicle's operational requirements. This could necessitate an extension of the time window for

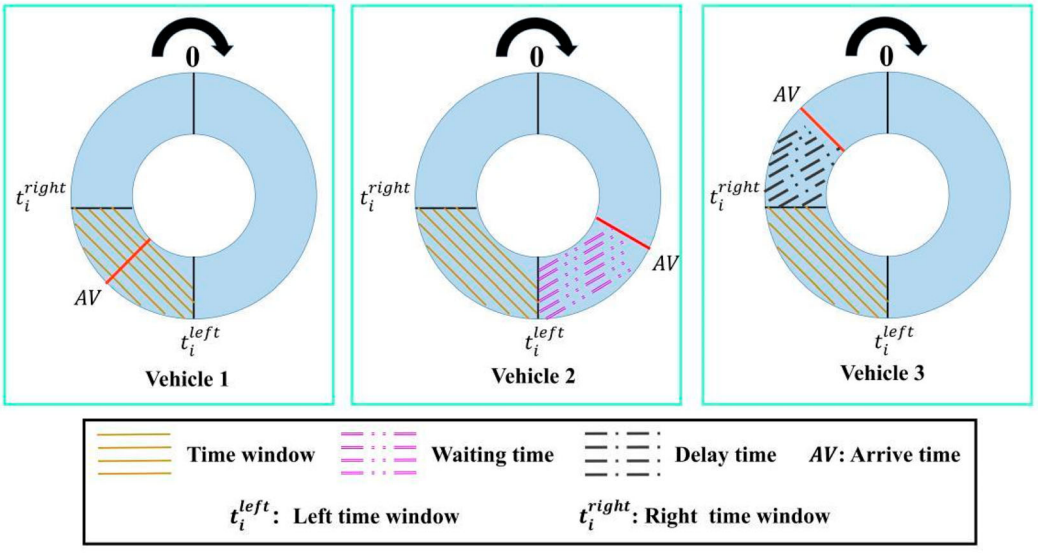


Figure 2. Description of time window violations.

the medical unit, which could adversely impact both the medical unit's operations and the vehicle's schedule. Vehicles 2 and 3 in the scenarios described incur penalties to varying extents.

Therefore, such penalties should be minimized.

The primary goal of this article is to optimize waste management within the context of the MWM system. It specifically targets the reduction of transportation-related costs and penalties incurred due to time window violations. An optimization model is introduced, designed to improve the efficiency and calibre of waste management operations, with the ultimate aim of reducing the total financial burden of the process.

Furthermore, the proposed model alleviates financial pressures on governmental budgets within relevant sectors. It also aims to streamline the waste collection infrastructure, ensuring a more coherent and effective network for managing MW. This comprehensive approach seeks to balance economic considerations with environmental protection and public health imperatives.

When the application scenario of the example is scaled up to a certain extent, the number of medical units that need to be considered is difficult to estimate. This optimization problem presents a considerable challenge owing to its classification as an NP-hard problem. Therefore, it is crucial to meticulously design test cases to address and evaluate the problem effectively.

According to the 14th Five-Year Health and Health Development Plan for Nanjing, there are 3439 hospital institutions in the city. However, including all scales of medical units would significantly increase the workload. Nanjing is used as a case study to simplify the problem and more effectively demonstrate the importance of the solution. It collects data on megawatt generation from several large medical units in the area. These representative sample data carry more persuasive weight when deriving the optimal solution.

Moreover, specific route planning for vehicles must consider TTR and various constraints. The collection of MW should start from disposal centres for each medical unit to ensure that every unit is serviced. Once the vehicles are fully loaded and have returned to the disposal centres, a viable network for urban MW collection will be established.

5.2. Model assumptions

To streamline the construction of the model, some necessary assumptions are given:

(1) Each medical unit is serviced once, whereas each vehicle can service multiple medical units (Hassanpour *et al.* 2023; Shen *et al.* 2024; Xu *et al.* 2023).
(2) The travel time for each link is assumed to follow a log-normal distribution (Shen *et al.* 2022).
(3) Multiple vehicles are permitted to depart at the same time, with their routes operating without mutual interference.
(4) The specifications and operational costs of the vehicles are based on realistic and practical considerations.
(5) The operational time windows for disposal centres are sufficiently broad to accommodate various scheduling scenarios.
(6) The transportation cost associated with waste collection is solely dependent on the quantity of waste and does not account for other potential factors.
(7) The specific processes involved in waste treatment at disposal centres are not taken into consideration.
(8) Disposal centres are assumed to have an unlimited processing capacity for the waste received.
(9) The mean and variance values for road travel times are based on realistic and empirical data.

5.3. Mathematical model

5.3.1. Symbols and notation

Table 2 provides a summary of the symbols and notation that are used throughout this article. This section provides a detailed analysis of the mathematical model formulated using the notation outlined in Table 2. This structured approach ensures clarity and coherence in developing and understanding the subsequent mathematical framework.

5.3.2. Model construction

The primary focus is on devising the most efficient driving routes for transportation vehicles. This strategic planning considers critical factors, including minimizing transportation costs and avoiding time window penalties. These different aspects will be characterized separately to ensure an efficient and effective route plan.

5.3.2.1. Transportation cost.

$$\min \left\{ \sum_{v\in V} \sum_{i\in D} y_i f_v + \sum_{v\in V} \sum_{i\in M\cup D} \sum_{j\in M\cup D} x_{vij} l_i d_{ij} ue \right\} \tag{4}$$

$$\sum_{v\in V} \sum_{i\in N} x_{vij} \geq 1 \forall j \in N, i \neq j \tag{5}$$

$$\sum_{v\in V} \sum_{j\in M} x_{vij} = 1 \forall i \in M \tag{6}$$

$$\sum_{j\in M} x_{v0j} = 1 \forall v \in V \tag{7}$$

$$\sum_{i\in D} x_{vij} - \sum_{i\in M} x_{vji} = 0 \forall v \in V, j \in N \tag{8}$$

$$\sum_{i\in D} x_{vi,n+1} = 1 \forall v \in V \tag{9}$$

$$\sum_{i\in N} x_{v0i} = \sum_{i\in N} x_{vi0} \forall v \in V \tag{10}$$

Equation (4) is designed to minimize the total transportation and vehicle usage costs, aiming to optimize the efficiency of the routing plan. Equation (5) enforces the constraint that every medical unit must be visited at least once, ensuring comprehensive coverage in the routing strategy. Equation (6) stipulates that each medical unit can only be assigned to a single route, preventing duplication of visits and maintaining the clarity of the routing assignments. Equations (7)–(9) are responsible for defining the flow constraints of the transportation vehicle along the route. Lastly, Equation (10) mandates that each transport formation must both start and end its journey at a disposal centre, thereby ensuring that the routing forms a closed loop and adheres to the operational guidelines.

5.3.2.2. Time window penalty.

$$min\left\{\sum_{i\in M}(\alpha\delta_i + \beta\epsilon_i)\right\} \tag{11}$$

$$\begin{cases} \delta_i = lt_{i-1} + t^{\theta}_{i-1,i} + ser_i - t_i^{right} & lt_{i-1} + t^{\theta}_{i-1,i} + ser_i > t_i^{right} \epsilon_i = t_i^{left} - lt_{i-1} - t^{\theta}_{i-1,i} \\ lt_{i-1} + t^{\theta}_{i-1,i} < t_i^{left} \delta_i = 0 \; and \; \epsilon_i = 0 & lt_{i-1} + t^{\theta}_{i-1,i} \geq t_i^{left}, \quad lt_{i-1} + t^{\theta}_{i-1,i} + ser_i \leq t_i^{right} \end{cases} \tag{12}$$

$$et_{i+1} \geq lt_i + ser_i + t^{\theta}_{i-1,i} \forall i \in M \cup D \tag{13}$$

Table 2. Symbols and descriptions.

Symbol	Description
Sets	
$G = (N, A)$	In a directed network G, $N = \{0, 1, 2, \cdots, n, n+1\}$ represents the set of all nodes, where 0 and $n+1$ represent the disposal centre and A represents the set of links. It is stipulated that for a directed graph G, a valid distribution route must originate from node 0 and terminate at node $n+1$
M	Set of medical units, $M \subseteq N \backslash \{0, n+1\}$
M^*	Set of medical units that have been served, $M^* \subseteq N \backslash \{0, n+1\}$
D	Set of disposal centres
V	Set of vehicles, $v \in V$
R^*	Set of positive rational numbers
Parameters	
d_{ij}	Distance between two random nodes
θ	Reliability level
w_i	The amount of medical waste for medical unit i
$[t_i^{left}, t_i^{right}]$	Soft time window for medical unit i
ser_i	Service time of medical unit i
l^v_{max}	Maximum loading capacity of vehicle v
l_i	Loading capacity of the vehicle at the mth medical unit i
ue	Transport cost parameter: represents the unit expense incurred for every unit of weight of waste transported over a specified distance
f_v	Fixed cost of using vehicle v
α	Penalty coefficient for lateness at medical unit i
β	Penalty coefficient for earliness at medical unit i
t^{θ}_{ij}	Travel time on arc (i, j), under reliability level θ
Variables	
x_{vij}	$x_{vij} = 1$ denotes that vehicle v visits j from i, and $x_{vij} = 0$ otherwise
δ_i	Expected lateness
ϵ_i	Expected earliness
et_i	Arrival time at medical unit i
lt_i	Departure time from medical unit i

$$\{t_0^{left} \equiv t_{n+1}^{left} = -\infty t_0^{right} \equiv t_{n+1}^{right} = +\infty \tag{14}$$

$$x_{vij} \in \{0, 1\} \tag{15}$$

$$et_i \in R^* \tag{16}$$

$$lt_i \in R^* \tag{17}$$

$$\delta_i \in R^* \tag{18}$$

$$\epsilon_i \in R^* \tag{19}$$

Equation (11) is formulated to minimize the penalty associated with time window violations. Equation (12) serves a multifaceted role, calculating the arrival time at each medical unit and the departure time from the preceding medical unit, and evaluating the expected lateness and earliness times for each stop. Equation (13) imposes a logical sequence constraint, dictating that the arrival time at any given medical unit must not precede the sum of the arrival time at the previous unit, the service time at that unit and the transportation time along the route segment between the two units $(i-1, i)$. Lastly, Equation (14) specifies the time window for the disposal centre, setting it to zero, which implies that the disposal centre operates on a continuous schedule without specific time window restrictions.

5.3.3. Multi-objective transformation

Section 5.3.2 presents the model construction, focusing on the optimal path planning for transportation vehicles while considering transportation costs and time window penalties. Typically, solutions to multi-objective optimization problems aim for Pareto optimality and are attained through exact or heuristic algorithms. Therefore, the optimal solution proposed in this article should account for path planning considering specific transportation costs and time penalties. In real-world scenarios, the focus often lies on the variability of total costs measured in monetary terms. This study construed the time window penalty coefficient as the penalty incurred due to timeouts or waiting. Equation (20) represents the simple arithmetic addition of the two objectives, deviating from the weighted sum method typically used for multi-objective problems (owing to the solution space of this model not meeting convexity requirements). To summarize, the optimization model for CVRPTW can be articulated as follows:

Objective:

$$min \left\{ \sum_{v \in V} \sum_{i \in D} y_i f_v + \sum_{v \in V} \sum_{i \in M \cup D} \sum_{j \in M \cup D} x_{vij} l_i d_{ij} ue + \sum_{i \in M} (\alpha \delta_i + \beta \epsilon_i) \right\} \tag{20}$$

subject to constraints (5)–(10) and (12)–(19).

6. Solution algorithm

Obviously, the proposed model is a VRP problem and many existing heuristic algorithms can effectively solve such problem (Cen *et al.* 2023; Pasha *et al.* 2022). Given the large scale of the problem, this article presents an IHSGA, designed to address the proposed optimization model. The reliable path, represented by effective travel time, is based on the research conducted by Shen *et al.* (2019, 2020, 2022). Section 7, on effectiveness, demonstrates that the proposed algorithm surpasses general heuristic algorithms regarding solution quality.

6.1. Calculation of path effective travel time

Assuming that the travel time for a given road segment follows a log-normal distribution, the method of moments-matching (MOM) is applied. This approach is consistent with the technique previously

used by Yang and Gao (2020). The MOM is a reliable statistical tool for estimating the distribution parameters, allowing for a more accurate representation of the travel time variability. First, the moments M_r and central moment D_r are calculated as follows:

$$M_r = \sum_{r \in R} exp(\mu_{ij}^2 + 0.5\sigma_{ij}^2)x_{vij} \tag{21}$$

$$D_r = \sum_{r \in R} \mathrm{cov}(t_{ij}^{\theta}, t_{mn}^{\theta})x_{vij}x_{vmn} = \sum_{r \in R} exp(2\mu_{ij}^2 + \sigma_{ij}^{2^2})(exp(\sigma_{ij}^{2^2}) - 1)x_{vij} + \sum_{r \in R} x_{vij}x_{vmn} \times \sqrt{exp(2\mu_{ij}^2 + \sigma_{ij}^{2^2})(exp(\sigma_{ij}^{2^2}) - 1)exp(2\mu_{mn}^2 + \sigma_{mn}^{2^2})(exp(\sigma_{mn}^{2^2}) - 1)} \tag{22}$$

Now, the mean and variance of the travel time of the entire path r within the path set R can be calculated:

$$\sigma_r^2 = \sqrt{LN(1 + D_r/M_r^2)} \tag{23}$$

$$\mu_r^2 = LN(M_r) - 0.5\sigma_r^{22} \tag{24}$$

Secondly, given the non-additivity of the log-normal distribution, the travel time of the path r cannot be calculated directly. The following scale is used for processing:

$$exp(\mu_r^2) \le T_r \le exp(\mu_r^2 + \Phi^{-1}(\theta)\sigma_r^2) \tag{25}$$

Expressing Equation (24) in $exp(\mu_r^2)$, the following formula is given:

$$exp(\mu_r^2) = exp(LN(M_r) - 0.5\sigma_r^{2^2}) = M_r * exp(-0.5\sigma_r^{2^2}) \tag{26}$$

According to Srinivasan, Prakash, and Seshadri (2014) and Yang and Gao (2020), a lower bound of effective travel time can be given:

$$T_r \ge exp(\mu_r^2) = exp(LN(M_r - 0.5\sigma_r^{2^2}) = M_r * exp(-0.5\sigma_r^{2^2}) \ge M_r * exp(-0.5\sigma_{\max}^2) \tag{27}$$

Therefore, the formula can realign the lower bound of effective travel time:

$$T_{r,\min} = M_r * exp(-0.5\sigma_{\max}^2) \tag{28}$$

Thus, a lower boundary for the travel times of the paths has been delineated, which is then used as an input parameter for the subsequent algorithm. This algorithm is designed to devise the corresponding feasible short circuits. Within the context of this process, the output yielded by the subsequent algorithm is recognized as indicative of the actual path travel time. This systematic approach ensures a structured method for planning and optimizing the routing strategies.

6.2. Improved harmony search genetic algorithm (IHSGA)

Since their introduction by American scholar John H. Holland in 1975, GAs have been extensively applied to many variations and formulations of the VRP and its numerous extensions. Nevertheless, comparative study reveals that incorporating specific parameters—namely, the harmony memory considered rate (HMCR), pitch adjustment rate (PAR) and bandwidth (BW)—from the harmony search (HS) algorithm can significantly enhance the diversity of newly generated solutions. This integration with the HS algorithm presents a promising avenue for enhancing the exploration capabilities of GA, potentially leading to more robust and varied solution sets in the context of VRP and its

variants. This approach offers a higher likelihood of escaping local optima and operates more efficiently than traditional GA. Building on these insights, this article develops an adaptive strategy that dynamically adjusts the parameters. Consequently, it replaces the conventional crossover and mutation operations of GA with a novel solution generation method that combines adaptive global search with harmony search techniques (IHSGA).

The specific steps in the IHSGA are as follows.

Step 1. Encoding and decoding
A critical aspect of applying IHSGA to the CVRPTW is the method of chromosome encoding. Using a concise encoding approach can enhance the algorithm's efficiency. This study employs an integer coding scheme where the processing centre and medical units are encoded within the chromosome. For instance, with a chromosome length determined by the sum of medical units and the maximum number of vehicles minus one (*e.g.* 8), a route that includes processing centre 1 followed by medical units 4, 2 and 3 would be encoded as (1, 4, 2, 3, 5, 6, 7, 8).

Step 2. Fitness function
The encoding strategy previously described does not automatically ensure compliance with the prescribed load capacity and time window constraints for all decoded delivery routes. To rectify this limitation, a penalty function is applied. This function adjusts the delivery plan's total cost, ensuring that routes violating the constraints are penalized accordingly. The revised total cost is computed using the formula (20). This approach aims to incentivize the generation of delivery routes that are both feasible and efficient, aligning with the operational requirements and constraints of the system.

$$f(s) = c(s) + \alpha \times q(s) + \beta \times \Omega(s) \tag{29}$$

$$q(s) = \sum_{r_i=1}^{\Omega} \max\left\{\left(\sum_{i\in M} w_i^{\theta} \sum_{v\in V}\sum_{i\in N}\sum_{j\in N} x_{vij} - C\right), 0\right\} \tag{30}$$

$$w(s) = \sum_{i=1}^{n} \max\{(lt_i - t_i^{right}), 0\} \tag{31}$$

where $c(s)$ is the transportation cost, $q(s)$ is the capacity constraint penalty cost, and $\Omega(s)$ is the time window penalty cost.

Step 3. Population initialization
Before the commencement of the population for the CVRPTW, it is essential to devise an initial solution framework. It is worth noting that this preliminary solution may not invariably conform to the load capacity and time window stipulations. Nevertheless, formulating a carefully constructed initial solution can substantially simplify the complexity inherent in the search process within GA. This strategic approach lays the groundwork for a more efficient solution for space exploration, facilitating a smoother optimization trajectory.

Step 4. Binary tournament selection operation
Binary tournament selection is a process that involves the comparison of pairs of individuals, with the individual possessing the higher fitness value being chosen for the subsequent generation. This selection cycle is iterated several times, equivalent to the size of the population. During each iteration, a pair of individuals is randomly selected for assessment, and the individual demonstrating superior fitness is advanced to the offspring population. It is worth noting that this method may lead to the inclusion of duplicate individuals within the selected pool. In cases where duplicates occur, a protocol is followed to retain only a single instance of the duplicate individual, with the remaining duplicates

being systematically eliminated. This ensures a diverse and non-repetitive composition within the offspring population.

Step 5. Chromosome adjustment (formation of new harmony)
In the conventional framework of GA, the selection phase is typically succeeded by crossover and mutation operations. However, this study deviates from the norm by replacing the crossover and mutation steps with an alternative approach. This novel method is designed to generate new solutions that may either fall within or transcend the boundaries of the harmony memory (HM). The generation of these solutions is guided by a set of rules that are influenced by parameters such as HMCR, PAR and BW, all of which are integral to the HS algorithm. This paradigm shift aims to integrate the exploratory and exploitative capabilities of the HS algorithm, thereby enhancing the solution search process within the GA.

Ultimately, the process culminates in the acquisition of a novel chromosome. Furthermore, as the iterative process advances, certain parameters, denoted by PAR and BW, will undergo modification. These alterations occur in tandem with the progression of the iteration count, as detailed in the following equations:

$$PAR(t) = PAR_{\min} + \frac{PAR_{\max} - PAR_{\min}}{NI} \times t \tag{32}$$

$$BW(t) = BW_{\max} exp\left(\frac{In\left(\frac{BW_{\min}}{BW_{\max}}\right)}{NI}\right) \times t \tag{33}$$

Where $PAR_{\min}$, $PAR_{\min}$, $BW_{\max}$ and $BW_{\min}$ are preset.

The specific code for this step is as follows:

Set the parameters *HMS*, *HMCR*, *PAR*, *BW*, $T_{\max}$
for ($j = 1$ to n) *do*
  Let p_1 be a random value within the interval (0, 1)
  if ($p_1 < HMCR$) *then*
    $X_{new}(j) = X_a(j)$ where $a \in (1, 2, \ldots, NIND)$
    Let p_2 be a random value within the interval (0, 1)
    if ($p_2 < PAR$) *then*
      Let p_3 be a random value within the interval (0, 1)
      $X_{new}(j) = X_{new}(j) \pm p_3 \times BW$
    endif
  else
    Let p_4 be a random value within the interval (0, 1)
    $X_{new}(j) = LB_j + p_4 \times (UB_j - LB_j)$
  endif
endfor
OutputX_{new}

Step 6. Local search operation (LSO)
The LSO used in this study is inspired by the foundational principles of destruction and repair that are hallmarks of large-scale neighbourhood search techniques. In detail, the destruction phase of the operator commences by methodically eliminating a specific number of customers from the current solution. Subsequently, the repair phase takes over, meticulously reintroducing the removed customers into the solution to form a revised arrangement. This two-step process enhances the solution by strategically disrupting and refining the existing configuration.

It is crucial to emphasize that the destruction operator does not engage in random customer removal. Instead, it employs a similarity calculation formula to identify and eliminate multiple customers with high similarity. Likewise, the repair operator does not insert the removed customers

Table 3. Relevant data for the random example.

Serial no.	t_i^{left}	t_i^{right}
1	0	∞
6	50	52
8	58	60

into arbitrary positions within a path. On the contrary, it strategically repositions the customers in a manner that adheres to the capacity and time window constraints while simultaneously aiming to minimize the incremental increase in the vehicle's travel distance. This targeted approach ensures a more refined and efficient search within the solution space.

Step 7. Recombination operation
During the execution of binary tournament selection, retaining only a single instance of any duplicate individuals is imperative, with the remainder being discarded. This practice is essential to ensure that the total count of the selected offspring does not surpass the size of the original population. Local search operations are strategically applied to the offspring to maintain the population size, followed by a recombination phase. This phase is pivotal as it integrates the post-local search offspring with the initial population from the start of the current iteration, thus preserving the consistent size of the population.

Given that the population size is denoted as *NIND* and the number of individuals selected through binary tournament selection is designated as *NIND*_1, the recombination operation adheres to a specific principle. Specifically, it involves retaining all *NIND*_1 offspring individuals in the post-local search from the current iteration. It also involves retaining only those individuals from the original population at the start of the current iteration who possess the highest fitness values ($NIND - NIND_1$). This methodical approach ensures the preservation of the fittest while also allowing for the infusion of new genetic material into the population. Figure 3 shows the summarized algorithm flow.

7. Numerical examples

7.1. Simple traffic network

To validate the efficacy of the algorithm and the feasibility of the model, a random example is presented to compare the solutions derived from IHSGA with the optimal solution attained through enumeration and a generic solver (CPLEX).

In this example, a rectangular network comprises eight nodes interconnected by 11 links. The travel times across each road segment are subject to a normal distribution, defined by distinct parameters that capture the variability in travel times. The Appendix contains Tables A1 and A2, which delineate the mean and variance of the travel times for each segment, respectively. Within this scenario, a designated node assumes the role of the disposal centre, equipped with a single vehicle for waste management. Concurrently, two nodes are randomly selected to represent medical units. Table 3 outlines the basic details for these three nodes in a manner that maintains general applicability. For further clarity, Figure 4 provides a schematic representation of the intricate traffic network under consideration. This example is based on TTR, and TTR is set to be 80%.

First, the travel time between any two adjacent points within the network is calculated using the formulae in Section 6.2, and the results are shown in Figure 5.

Based on the data presented in Figure 5, an enumeration is conducted to calculate the associated costs for all viable transportation schemes. A comprehensive compilation of these results is provided in Table 4. It is noteworthy that the overall travel time for the route established by IHSGA is recorded as 80.5705, with the specific route being $1 \rightarrow 3 \rightarrow 6 \rightarrow 8 \rightarrow 5 \rightarrow 2 \rightarrow 1$. Upon examination of Table 4, it becomes clear that, out of all potential transportation routes, only routes 6–15 successfully meet

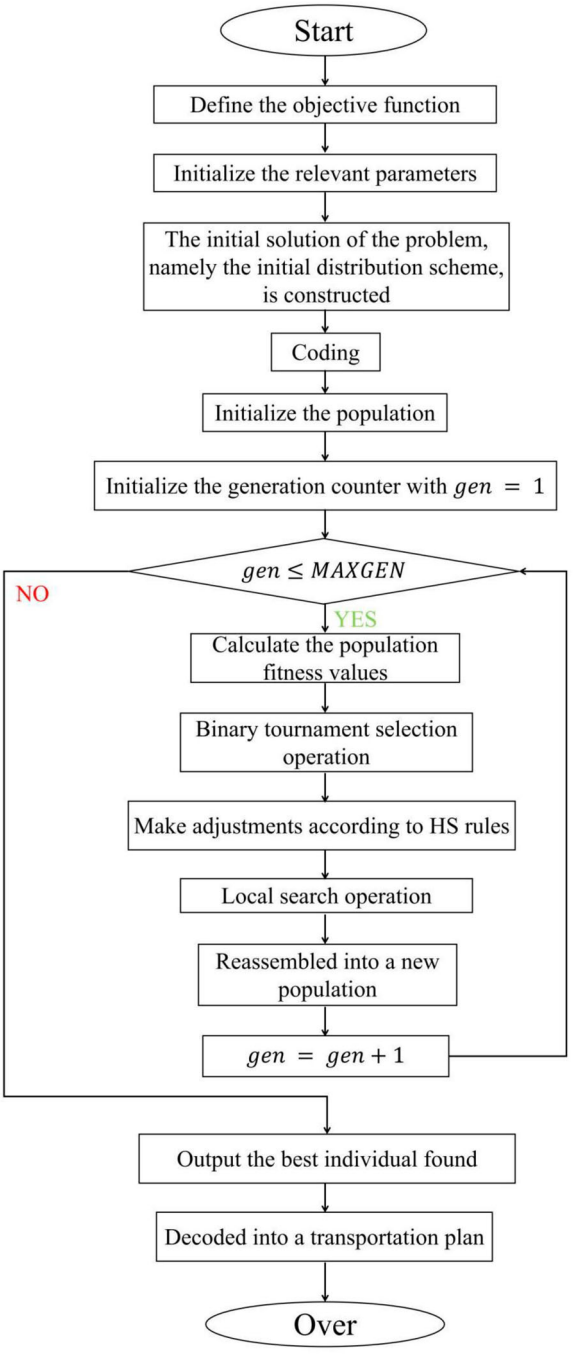


Figure 3. Flowchart of the proposed improved harmony search genetic algorithm (IHSGA). SA = ; HS = harmony search.

the time window constraints for each medical unit. Furthermore, among these feasible routes, route 6 is distinguished by having the shortest travel time, which corroborates the findings of IHSGA.

Following a thorough comparison between the outcomes achieved through enumeration and those produced by IHSGA, CPLEX was used to reassess the example. As illustrated in Figure 6, when the

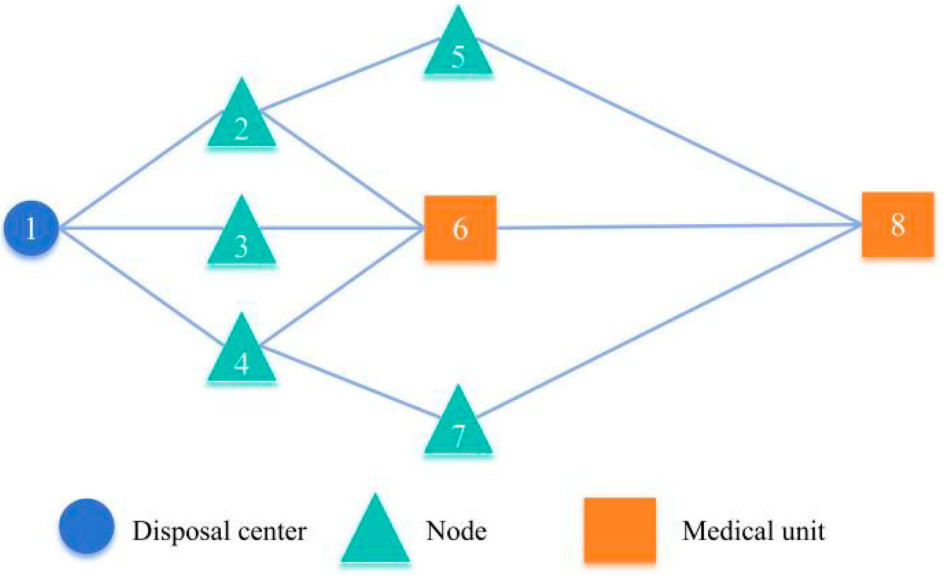


Figure 4. Enumeration example diagram.

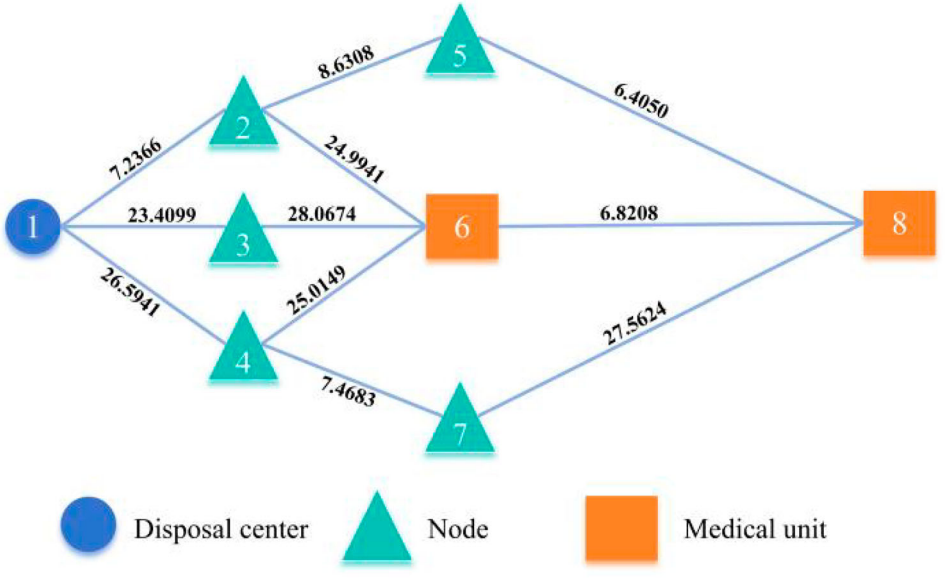


Figure 5. Travel time of each section.

generic solver's initial solution value is 116.728, the optimal value of 80.5705 is reached in the seventh iteration after 2.3 min of processing. This result aligns with the calculations from the other two methods, affirming the validity of IHSGA and providing robust validation for the model. Therefore, it is justifiable to think that IHSGA can solve CVRPTW well.

Table 4. Enumeration results.

Serial number of route	Transportation route	AT_1	AT_2	Total time	Does the route violate the time window?
1	1 → 2 → 6 → 8 → 5 → 2 → 1	32.2307	39.0515	61.3239	×
2	1 → 2 → 6 → 8 → 6 → 2 → 1	32.2307	39.0515	78.1030	×
3	1 → 2 → 6 → 8 → 6 → 3 → 1	32.2307	39.0515	97.3496	×
4	1 → 2 → 6 → 8 → 6 → 4 → 1	32.2307	39.0515	97.4813	×
5	1 → 2 → 6 → 8 → 7 → 4 → 1	32.2307	39.0515	100.6763	×
6	1 → 3 → 6 → 8 → 5 → 2 → 1	51.4773	58.2981	80.5705	✓
7	1 → 3 → 6 → 8 → 6 → 2 → 1	51.4773	58.2981	97.3496	✓
8	1 → 3 → 6 → 8 → 6 → 3 → 1	51.4773	58.2981	116.5962	✓
9	1 → 3 → 6 → 8 → 6 → 4 → 1	51.4773	58.2981	116.7279	✓
10	1 → 3 → 6 → 8 → 7 → 4 → 1	51.4773	58.2981	119.9229	✓
11	1 → 4 → 6 → 8 → 5 → 2 → 1	51.6090	58.4298	80.7022	✓
12	1 → 4 → 6 → 8 → 6 → 2 → 1	51.6090	58.4298	97.4813	✓
13	1 → 4 → 6 → 8 → 6 → 3 → 1	51.6090	58.4298	116.7279	✓
14	1 → 4 → 6 → 8 → 6 → 4 → 1	51.6090	58.4298	116.8596	✓
15	1 → 4 → 6 → 8 → 7 → 4 → 1	51.6090	58.4298	120.0546	✓
16	1 → 2 → 5 → 8 → 6 → 2 → 1	22.2724	29.0932	61.3239	×
17	1 → 2 → 5 → 8 → 6 → 3 → 1	22.2724	29.0932	80.5705	×
18	1 → 2 → 5 → 8 → 6 → 4 → 1	22.2724	29.0932	80.7022	×
19	1 → 4 → 7 → 9 → 6 → 2 → 1	61.6248	68.4456	100.6763	×
20	1 → 4 → 7 → 9 → 6 → 3 → 1	61.6248	68.4456	119.9229	×
21	1 → 4 → 7 → 9 → 6 → 4 → 1	61.6248	68.4456	120.0546	×

Note: AT_1 = time when the vehicle arrives at the first destination; AT_2 = time when the vehicle arrives at the second destination.

Table 5. Benchmark example data set.

				Scenario 1		Scenario 2		Scenario 3		
Customer no.	X	Y	w_i	L	R	L	R	L	R	Service time
1	40	50	0	0	240	0	240	0	240	0
2	25	85	20	71	191	69	79	45	165	10
3	22	75	30	30	150	32	152	94	119	10
4	22	85	10	64	184	152	162	75	85	10
5	20	80	40	151	161	35	117	128	194	10
6	20	85	20	40	160	68	78	171	181	10
7	18	75	20	96	123	59	114	60	93	10
8	15	75	20	35	155	147	180	147	180	10
9	15	80	10	101	111	79	124	55	120	10
10	10	35	20	101	111	58	115	52	112	10
11	10	40	30	123	144	111	162	75	102	10

Note: L = left time window; R = right time window.

7.2. Benchmark instance

In Section 7.1, a small-scale example was used to verify the effectiveness of the model and the algorithm. Before moving on to real-world scenarios, it is worth noting that the time window for each medical unit is typically fixed. However, it is valuable to investigate whether altering the time window could cause the algorithm to fail in specific cases. Hence, this section examines the impact of time window variations on path planning using a benchmark example from the Solomon RC105 data set (web.cba.neu.edu/ ∼ msolomon/rc101.htm). This study focuses on path planning when the geographical positions of medical units remain constant but the time window changes (progressing through Scenarios 1, and 3, sequentially). Table 5 presents the data for the selected sample, while Table 6 illustrates the path-planning outcomes after the time window of the same medical unit is altered twice.

Table 6 demonstrates that when medical units at the same location are assigned different time windows, their planned paths also vary. This outcome aligns with expectations in real-world scenarios:

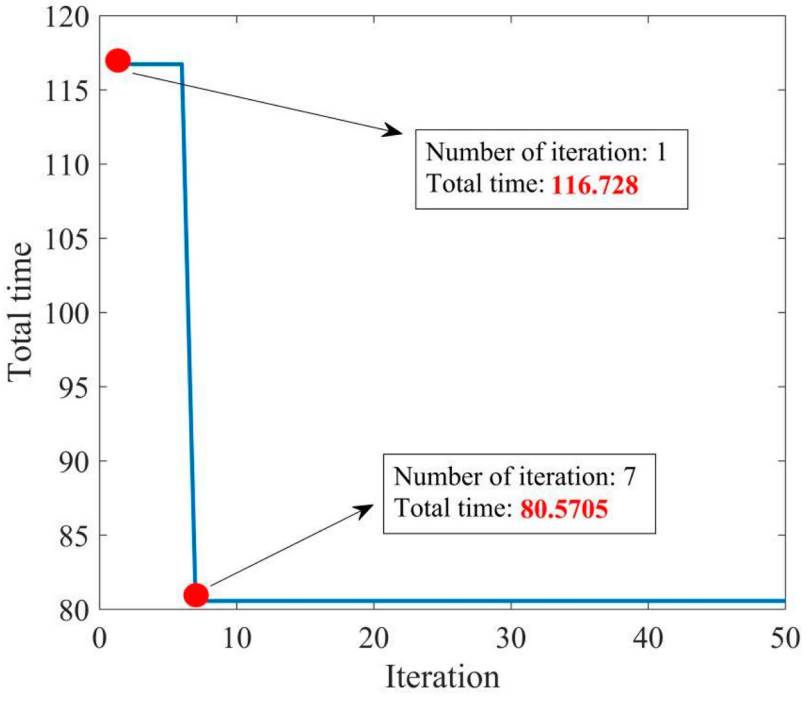


Figure 6. Plot computed by the generic solver.

Table 6. Calculation results for different scenarios.

Scenario	Path planning	NP	NC
1	Route 1: 0 → 1 → 3 → 5 → 4 → 0 Route 2: 0 → 9 → 10 → 0 Route 3: 0 → 7 → 8 → 6 → 2 → 0	0	0
2	Route 1: 0 → 1 → 6 → 2 → 7 → 0 Route 2: 0 → 5 → 4 → 8 → 3 → 0 Route 3: 0 → 9 → 10 → 0	1	4
3	Route 1: 0 → 6 → 8 → 2 → 5 → 0 Route 2: 0 → 3 → 1 → 4 → 7 → 0 Route 3: 0 → 10 → 9 → 0	1	4

Note: NP = number of path constraint violations; NC = number of customer constraint violations.

altering established routes can incur additional costs that are often not cost-effective. In conclusion, the algorithm effectively addresses situations involving changes in time windows, showcasing its capability to adapt and optimize path planning under varying temporal constraints.

7.3. Example construction

Figure 7 displays a portion of Q district in N city, highlighting eight medical units (consisting of large hospitals and clinics graded III or higher) and one waste disposal centre.

The numerical experiments were conducted using MATLAB® 2021b, operating on a Windows 11 platform. The hardware specifications included an Intel Core™ i5-11260H 2.6 GHz CPU and a system memory of 16 GB. The quantities of waste generated, along with the mean and variance of travel times for each road segment, were modelled to follow a log-normal distribution. These distributional data were randomly synthesized using MATLAB's built-in functions. The cost of transporting one

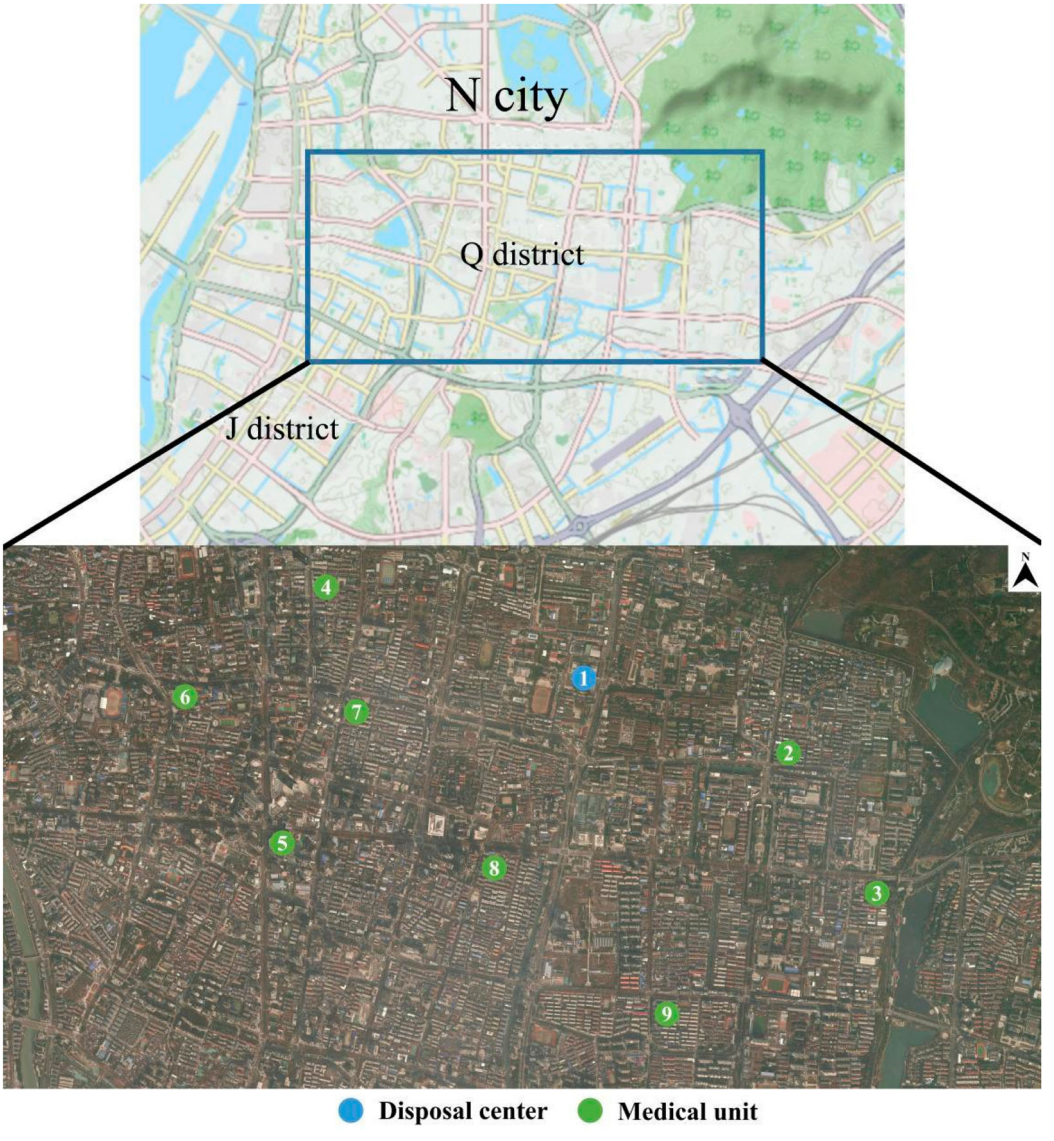


Figure 7. Reference diagram of Q district in N city.

kilogram of waste over one kilometre is expressed in US dollars. The operational parameters for the disposal centre include a vehicle cost and a load vector, which are represented as (250, 5000). These values are provided in the respective units of US dollars and kilograms, offering a clear understanding of the disposal centre's resource allocation and logistical considerations.

7.3.1. Calculation results

To provide a clear illustration, IHSGA was used to showcase the effects of varying values on the outcomes. Detailed results can be found in Table 7.

Some notable verdicts can be made by comparing the results for different TTR levels. When TTR is not considered, the total cost is 600.02; when TTR is 0.9, the total cost increases to 679.38. These results suggest that TTR affects the general operational costs of the waste recycling network.

Table 7. Results without and with consideration of travel time reliability (TTR).

Scenario	TTR	Total travel time (h)	Total distance (km)	Cost (US dollars)		
				Total cost	Cost of time window violation	Cost of transportation
Without consideration of TTR	$\theta = 0.5$	3.26	50.26	600.02	50.48	549.54
With consideration of TTR	$\theta = 0.6$	3.35	51.34	615.91	58.83	557.08
	$\theta = 0.7$	3.47	54.78	632.76	74.29	558.47
	$\theta = 0.8$	3.52	57.22	656.30	89.15	567.15
	$\theta = 0.9$	3.61	59.13	679.38	103.74	575.64

The itemized costs can also be reflected conclusions of practical significance. Regardless of the value of reliability compared to not considering reliability, the travel cost increases significantly, rapidly increasing from 549.54 ($\theta = 0.5$) to 575.64 ($\theta = 0.9$), and the same is true for the cost of time window violation, increasing from 50.48 ($\theta = 0.5$) to 103.74 ($\theta = 0.9$). Equation (1) supports this view: the higher the reliability level, the greater the equation value. Therefore, the results in Table 7 are reasonable.

In practical scenarios, an enhancement in TTR typically translates to an extended, reliable travel time. However, operational constraints at the disposal centre often prevent managers from initiating their MW collection operations before the designated time window. Consequently, the travel time required may surpass the soft time windows stipulated for medical units. This can lead to additional costs due to time window violations, a scenario that aligns with real-world operational expectations.

Figure 8 shows the detailed routing with and without TTR ($\theta = 0.5$ and $\theta = 0.9$, respectively), and the conclusions reflected therein seem to be able to illustrate expectations.

The differences between the two case paths demonstrate that the route that does not take TTR into account is shorter than the one that does. However, this does not indicate that the former is the optimal choice for waste transportation. Rather, factoring in TTR prompts managers to consider other unforeseeable factors, such as traffic congestion. In most cases, determining an optimal route requires weighing various considerations, including route length, traffic flow and potential disruptions. Although accounting for TTR may entail higher costs, it generally results in a more feasible solution with greater certainty than can be obtained when it is not factored in.

7.3.2. Sensitivity analysis

7.3.2.1. Effects of TTR on costs. In Section 7.3.1, cost comparisons were made across different levels of reliability, leading to specific conclusions. However, a detailed investigation into the cost variations arising from minor adjustments in reliability levels, using smaller increments, was deemed essential. To achieve this, an increment of 0.01 was implemented to meticulously analyse and monitor the cost changes while transitioning the reliability level from 0.5 to 0.99. The outcomes of this comprehensive analysis are depicted in Figure 9.

Figure 9 demonstrates that, as the TTR level varies ($\theta > 0.5$), it significantly impacts path planning. The three costs in the figure begin to fluctuate notably once the reliability level exceeds 0.5, showcasing nonlinear trends. Particularly noteworthy is the heightened variability in time window penalty costs compared to total and transportation costs. This discrepancy can be attributed to substantial shifts in transportation routes resulting from changes in reliability levels. While these alterations may not drastically impact travel distances, they can influence the occurrence of violated time windows owing to changes in the order of visits to medical facilities. Consequently, the ensuing additional costs may escalate, reflecting real-world scenarios.

7.3.2.2. Effects of penalty coefficients on total costs and routes. Adjusting the penalty coefficients for early and late arrivals can impact costs and routes. It is worth noting that the results presented in Table 7 were obtained with $\alpha = 0.2$ and $\beta = 0.5$. Table 8 displays the corresponding calculation results when these coefficients are varied within the range of [0.2, 0.5].

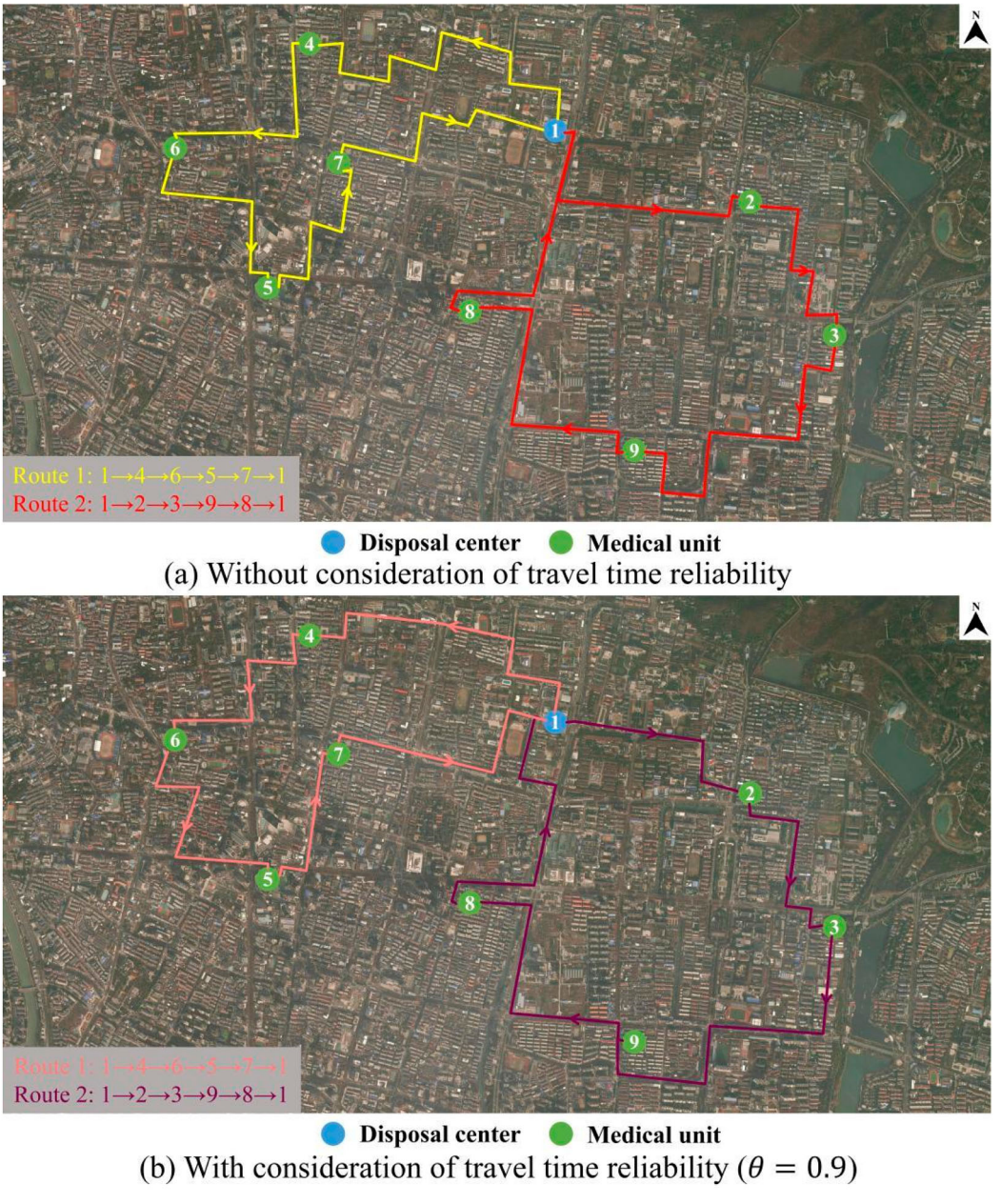


Figure 8. Detailed routing diagram for two cases: (a) without consideration of travel time reliability; (b) with consideration of travel time reliability ($\theta = 0.9$).

For Scenarios 1–5, it is evident that the difference between total and sub-costs is minimal, with a gap of less than 5%. This observation highlights the robustness of the model, indicating that artificially assigned penalty coefficients and variations in other parameters have negligible impact on the model itself. However, it is worth noting that when the two penalty coefficients are set at the same level (Scenarios 5 and 6), the associated costs for each scenario differ significantly from Scenarios 1–5.

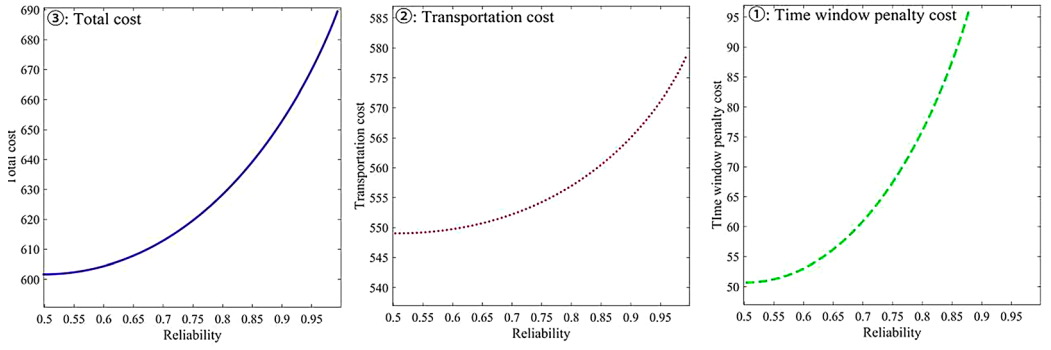


Figure 9. Sensitivity analysis of travel time reliability (TTR).

Table 8. Results after alpha and beta are changed.

Scenario	Total travel time (h)	Total distance (km)	Cost (US dollars)			Route 1	Route 2
			Total cost	Cost of time window violation	Cost of transportation		
(1) $\alpha = 0.2, \beta = 0.5$	3.61	59.13	679.38	103.74	575.64	$1 \to 4 \to 6 \to 5 \to 7 \to 1$	$1 \to 2 \to 3 \to 9 \to 8 \to 1$
(2) $\alpha = 0.3, \beta = 0.4$	3.68	58.97	677.06	106.88	570.18	$1 \to 4 \to 6 \to 5 \to 7 \to 1$	$1 \to 2 \to 3 \to 9 \to 8 \to 1$
(3) $\alpha = 0.4, \beta = 0.3$	3.57	59.06	676.26	101.16	575.10	$1 \to 4 \to 6 \to 5 \to 7 \to 1$	$1 \to 2 \to 3 \to 9 \to 8 \to 1$
(4) $\alpha = 0.5, \beta = 0.2$	3.73	54.77	674.89	114.58	560.31	$1 \to 4 \to 6 \to 5 \to 7 \to 1$	$1 \to 2 \to 3 \to 9 \to 8 \to 1$
(5) $\alpha = 0.2, \beta = 0.2$	3.82	48.23	651.09	139.23	511.86	$1 \to 4 \to 7 \to 5 \to 6 \to 1$	$1 \to 2 \to 3 \to 9 \to 8 \to 1$
(6) $\alpha = 0.5, \beta = 0.5$	3.82	48.08	671.55	152.00	509.55	$1 \to 4 \to 7 \to 5 \to 6 \to 1$	$1 \to 2 \to 3 \to 9 \to 8 \to 1$

This should be direction of arrows.

Furthermore, it is worth mentioning that even in the path depicted in Figure 10 of Scenario 6, the path direction changes from $1 \to 4 \to 6 \to 5 \to 7 \to 1$ to $1 \to 4 \to 7 \to 5 \to 6 \to 1$. This demonstrates that managers may subjectively deviate from the allocated time window for each medical unit in practical path planning. Such deviations are reasonable since, based on the map, the distance from 4 to 6 is considerably greater than that from 4 to 7.

It is crucial to recognize that the efficacy of heuristic algorithms is intricately linked to key parameters. As detailed in Section 6.2, IHSGA combines the GA and HS algorithm, meaning that its performance is influenced by parameters from both algorithms. In the context of HS, Pan *et al.* (2010) noted that an HMCR value equal to or greater than 0.9 was beneficial during algorithm refinement. Furthermore, a larger PAR value promotes the transfer of information from parent to offspring, expanding the search space and enriching harmonic diversity.

In the case of GA, Liu *et al.* (2023) successfully applied a GA with a trip-adjustment strategy to address the problem of multi-station electric bus scheduling, achieving favourable results with default parameters. This success suggests that default GA parameters could also play a positive role in enhancing the performance of IHSGA. However, it is worth noting that tools such as Irace offer efficient methods for automatic parameter tuning, offering valuable insights for future research pursuits.

7.3.3. Performance comparison among different algorithms

To assess the impact on operational costs, a comparative analysis was undertaken, in which TTR levels were systematically varied. The findings indicate a positive correlation between TTR and travel distance: as reliability increases, so does the distance travelled. This incremental distance, in turn, leads to an escalation in costs. This observation suggests that when TTR is increased, managers are more inclined to consider a broader range of uncertain factors during route planning. These results underscore the trade-offs between reliability and cost, offering valuable insights for strategic decision making in route optimization.

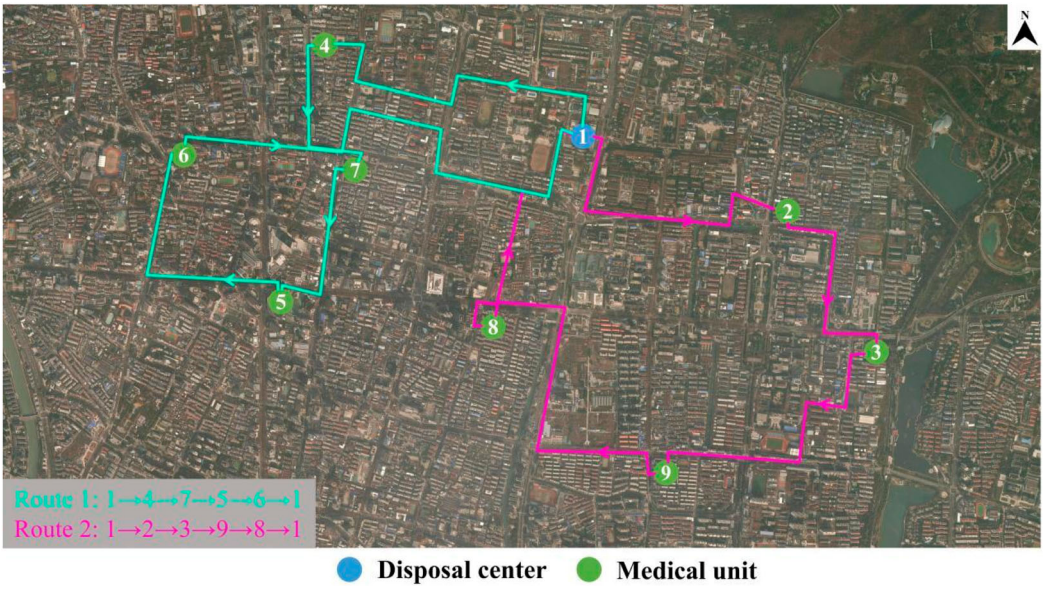


Figure 10. Detailed routing diagram of Scenario 6.

Table 9. Comparison of calculation outcomes for the genetic algorithm (GA), improved harmony search genetic algorithm (IHSGA) and variable neighbourhood search (VNS) examples.

	GA			IHSGA			VNS		
Trial run	Best (US dollar)	Worst (US dollar)	CPU (s)	Best (US dollar)	Worst (US dollar)	CPU (s)	Best (US dollar)	Worst (US dollar)	CPU (s)
1	725.24	879.38	100.97	681.26	743.97	31.98	685.50	846.37	39.50
2	732.28	927.37	100.13	687.30	791.61	32.78	686.89	811.45	37.60
3	724.66	992.61	101.86	679.38	857.12	32.49	689.32	789.02	40.84
4	733.33	851.62	101.57	700.01	715.64	31.68	677.29	773.91	40.58
5	731.01	937.05	101.67	675.29	801.61	31.37	701.60	800.32	36.03
Mean	729.304	917.606	101.24	684.648	781.99	32.06	688.12	804.214	38.91

IHSGA is anticipated to outperform the standard GA by integrating local search capabilities. However, it remains uncertain whether this proposed algorithm performs better than other heuristics that also use local search techniques, such as the iterative local search algorithm, variable neighbourhood search (VNS) or tabu search. A comparative analysis was conducted to assess the effectiveness of the algorithm, involving continuous computation outcomes and computation durations of IHSGA, GA and VNS across 50 successive runs. Detailed findings are available in Table A3 in the Appendix, while Table 9 summarizes five of these trial runs.

Regarding solution quality, when starting from the poorest solution with negligible disparities in the initial solutions, algorithms integrating local search functionalities such as IHSGA and VNS notably surpass those devoid of this feature, such as GA. Nevertheless, despite this observation, IHSGA exhibits only a slight, 0.5% edge over VNS, which does not notably boost the overall solution quality.

When examining solution time, GA exhibits a notable drawback in solution quality compared to local search algorithms such as IHSGA and VNS, with an average iteration time of 101.24, which is considered unsatisfactory. Allocating significant time for subpar results is often impractical. IHSGA shows a 17.60% decrease in iteration time (32.06) in contrast to VNS (42.91), despite no substantial variance in solution quality between the two. This validates IHSGA as offering both quicker solution times and enhanced solution quality. Figure 11 visually illustrates the iterative progression of the

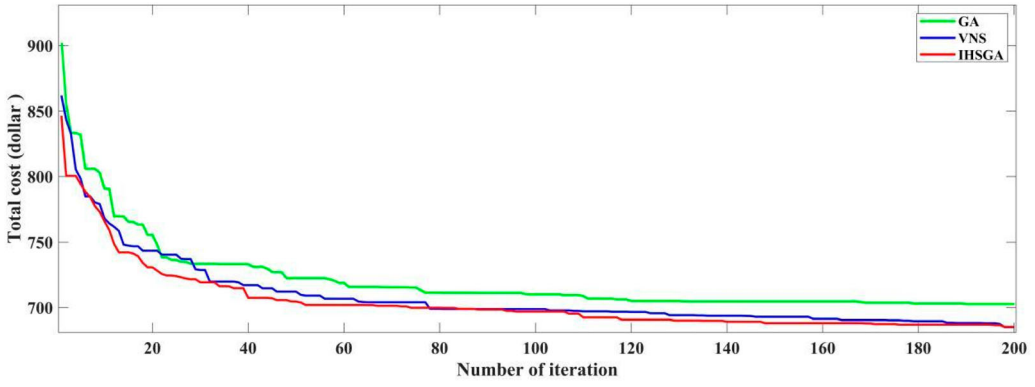


Figure 11. Plot depicting iterative outcome comparisons for the genetic algorithm (GA), improved harmony search genetic algorithm (IHSGA) and variable neighbourhood search (VNS).

three algorithms in a single run, further affirming the efficacy of IHSGA. These results emphasize the potential for integrating and optimizing diverse heuristics to improve performance.

Statistical tests are often the preferred method for determining whether differences exist between samples. While Table 9 and Figure 11 visually compare the differences among the three algorithms, it is necessary to verify statistically that IHSGA has advantages. According to the central limit theorem, when the sample size is sufficiently large (typically 50 or more), the sample distribution approximates a normal distribution. Therefore, this study used a single-sample analysis of variance (ANOVA) test to assess statistical differences among the three algorithms, with the results detailed in the Appendix.

The homogeneity of variance test results suggests that the sample data for solution quality among the three algorithms meet the assumption of homogeneity of variance. However, it is important to note that this assumption is not satisfied for solution time.

When assuming homogeneous variance for the samples of solution quality, various *post hoc* tests, such as the Tukey's HSD, Scheffé, least significant difference (LSD) and Bonferroni test, reveal significant differences between IHSGA and GA ($p < 0.05$). However, no significant difference is observed between IHSGA and VNS at the 0.05 significance level ($p < 0.05$). Dunnett's T3 test demonstrates that IHSGA differs significantly from GA and VNS regarding solution time. In summary, the results of the statistical tests align with the earlier conclusions, providing robust evidence supporting the superiority of IHSGA over GA in terms of solution quality, with comparable performance to VNS. This further validates the previous findings and emphasizes the strength of IHSGA in this context.

8. Conclusion

MWM is vital in urban operation and management, especially in densely populated areas. Poor management can lead to serious and unrecoverable economic, societal and ecological outcomes. Therefore, shifting away from conventional MW disposal approaches and prioritizing establishing sustainable MWM networks is imperative. Nevertheless, there are enormous challenges in building such networks resulting from the intricate processes and interconnections involved, especially in the context of global epidemics.

To address these issues, this study aims to account for the uncertainty related to the transportation of MW. It proposes a comprehensive model for designing MWM networks that incorporates transportation planning considerations. The IHSGA, based on adaptive and harmony search algorithms, is used to deal with this challenge. The key findings of this study can be summarized as follows:

(1) Integrating TTR into decision-making processes empowers decision makers to factor in additional considerations, albeit potentially resulting in increased costs attributed to prioritizing

reliability. Conversely, disregarding TTR could lead to cost reductions but may also bring about unforeseen consequences.

(2) Even minor adjustments in TTR can result in fluctuations in costs and routes, with a notable impact on time window penalties.

(3) IHSGA has notable advantages over the conventional GA regarding solution quality and speed. Compared to the VNS algorithm incorporating local search capabilities, IHSGA demonstrates comparable solution quality while surpassing it in solution speed. These distinctions were statistically verified through one-way ANOVA testing. Consequently, IHSGA exhibits the potential to optimize resource utilization, implement cost-saving strategies, and improve safety and efficiency in MWM, particularly given the inherent uncertainties associated with the process.

The findings of this study greatly contribute to mitigating decision risks in the MW collection problem, notably under uncertain conditions. In addition, they offer innovative insights for government-related personnel into effective management and operations.

Given the limitations identified in this study, the following suggestions for future research directions are proposed:

(1) To enhance resource integration, other stages of the recycling network could be considered.

(2) Alternative heuristic or multi-objective algorithms could be investigated to improve effectiveness and efficiency.

(3) In the realm of real-world MW transportation, attaining high-precision outcomes is paramount. Future initiatives will focus on developing precise algorithms to address small and medium-sized CVRPTW. These endeavours could provide decision makers with a more resilient framework for making well-informed decisions.

(4) Irace is widely acknowledged as a powerful tool for automatic parameter tuning, as highlighted in studies by Cáceres *et al.* (2017) and Ye *et al.* (2022). In future work, the intention is to use Irace to optimize the algorithm's performance, to achieve superior results through refined parameter settings.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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Data availability statement

The data that support the findings of this study are available from the corresponding author, Liang Shen, upon reasonable request.

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Appendix

Table A1. Mean travel time of each link.

Number	1	2	3	4	5	6	7	8
1	Inf	6.90	22.40	25.50	Inf	Inf	Inf	Inf
2	Inf	Inf	Inf	Inf	8.20	23.90	Inf	Inf
3	Inf	Inf	Inf	Inf	Inf	26.30	Inf	Inf
4	Inf	Inf	Inf	Inf	Inf	23.50	7.30	Inf
5	Inf	Inf	Inf	Inf	Inf	Inf	5.90	Inf
6	Inf	Inf	Inf	Inf	Inf	Inf	Inf	6.40
7	Inf	Inf	Inf	Inf	Inf	Inf	Inf	26.30
8	Inf	Inf	Inf	Inf	Inf	Inf	Inf	Inf

Table A2. Travel time variance of each link.

Number	1	2	3	4	5	6	7	8
1	Inf	0.16	1.44	1.69	Inf	Inf	Inf	Inf
2	Inf	Inf	Inf	Inf	0.25	1.69	Inf	Inf
3	Inf	Inf	Inf	Inf	Inf	4.41	Inf	Inf
4	Inf	Inf	Inf	Inf	Inf	3.24	0.04	Inf
5	Inf	Inf	Inf	Inf	Inf	Inf	0.36	Inf
6	Inf	Inf	Inf	Inf	Inf	Inf	Inf	0.25
7	Inf	Inf	Inf	Inf	Inf	Inf	Inf	2.25
8	Inf	Inf	Inf	Inf	Inf	Inf	Inf	Inf

Table A3. Comparison of calculation results among genetic algorithm (GA), improved harmony search genetic algorithm (IHSGA) and variable neighbourhood search (VNS) examples.

	GA		IHSGA		VNS	
Running time	Best (US dollar)	CPU (s)	Best (US dollar)	CPU (s)	Best (US dollar)	CPU (s)
1	725.24	100.97	681.26	31.98	685.50	39.50
2	732.28	100.13	687.30	32.78	682.99	39.19
3	724.66	101.86	679.38	32.49	687.47	36.17
4	733.33	101.57	700.01	31.68	681.92	36.34
5	731.01	101.67	675.29	31.37	686.89	37.60
6	721.38	101.65	678.44	31.64	681.86	38.65
7	721.12	101.11	685.77	32.18	683.71	39.27
8	727.78	100.78	682.88	32.02	686.28	38.04
9	722.16	100.93	682.58	32.31	687.83	40.10
10	727.85	101.08	678.53	32.25	680.84	39.59
11	721.25	101.26	685.61	32.43	689.32	40.84
12	727.80	100.93	683.30	32.42	687.78	38.66
13	728.17	101.27	680.59	31.86	684.89	37.63
14	719.93	101.38	682.21	32.23	684.39	36.53
15	719.75	100.89	681.10	31.73	684.49	39.05
16	722.08	100.79	677.84	31.56	683.09	39.89
17	727.37	100.97	679.48	33.07	685.11	38.12
18	720.64	100.99	678.31	32.83	686.79	36.45
19	722.10	100.25	678.92	32.81	689.86	37.33
20	720.33	100.34	679.48	33.23	689.63	36.77
21	727.38	99.92	687.29	32.94	688.13	37.41
22	720.84	100.09	683.62	32.95	685.47	38.20
23	721.35	100.63	692.15	33.19	689.80	38.64
24	728.24	99.86	692.57	33.14	687.01	38.29
25	727.64	100.76	688.03	32.91	685.19	40.38
26	727.06	100.56	688.01	32.22	691.08	38.59
27	735.13	100.32	686.50	32.28	690.44	40.72
28	726.99	100.41	692.12	32.93	687.19	39.19
29	726.94	100.07	686.81	32.07	682.89	40.79
30	727.63	102.02	676.31	32.53	682.54	37.20
31	734.67	102.52	683.00	32.21	678.74	39.38
32	727.55	102.11	679.10	33.02	679.68	37.45
33	734.42	102.08	677.62	32.75	681.38	39.36
34	727.60	101.79	679.24	32.54	678.97	39.48
35	733.73	102.05	676.16	32.51	685.11	36.34
36	727.97	102.18	676.52	32.10	678.61	37.27
37	728.03	102.24	684.62	32.72	678.93	37.12
38	722.39	101.96	684.76	32.08	678.37	39.34
39	729.24	101.93	690.58	31.30	678.94	40.22
40	723.46	102.55	685.43	31.75	681.02	37.72
41	722.24	101.60	687.18	31.33	700.41	39.90
42	729.03	102.45	688.36	32.05	706.53	39.38
43	736.31	102.47	693.04	32.05	701.60	36.03
44	729.58	102.07	684.98	31.95	699.15	39.01
45	734.70	101.37	685.26	31.07	706.35	37.93
46	729.03	101.53	686.52	31.58	677.29	40.58
47	733.36	101.61	691.32	31.44	674.88	36.01
48	733.76	101.95	692.15	31.89	681.02	38.31
49	726.74	101.41	691.31	31.57	674.52	38.12
50	733.93	101.99	675.62	31.72	674.84	38.30
Mean	727.2634	101.3064	684.0892	32.2332	685.5344	38.4476

Table A4. Statistical significance tests.

Project	Statistical test	(I) Group	(J) Group	*p*-Value
Solution quality	Tukey's HSD	IHSGA	GA	0.000
			VNS	0.446
	Scheffé	IHSGA	GA	0.000
			VNS	0.480
	LSD	IHSGA	GA	0.000
			VNS	0.226
	Bonferroni	IHSGA	GA	0.000
			VNS	0.679
Running time	Dunnett's T3	IHSGA	GA	0.000
			VNS	0.000

Note: LSD = least significant difference; IHSGA = improved harmony search genetic algorithm; GA = genetic algorithm; VNS = variable neighbourhood search.